Orthogonal projection operators

Definition. *A square matrix $P \in \mathbb{C}^{n\times n}$ is called an* orthogonal projection matrix *if it is Hermitian ($P^\dagger = P$) and satisfies $P^2 = P$.*

Let V be a k-dimensional subspace of $\mathbb{C}^n$, and $\{|u_1\rangle, \ldots, |u_k\rangle\}$ an orthonormal ($\langle u_i|u_j\rangle = \delta_{ij}$) basis of V (which always exists). Then

$$P = \sum_{j=1}^{k} |u_j\rangle\langle u_j|$$

(with $\langle u_j| = |u_j\rangle^\dagger$) is the *projector* onto V, acting on a vector $|w\rangle \in \mathbb{C}^n$ as

$$P|w\rangle = \sum_{j=1}^{k} |u_j\rangle\langle u_j|w\rangle.$$

It is straightforward to show that P is an orthogonal projection matrix, that P is independent of the chosen basis, and that $P|v\rangle = |v\rangle$ for all $|v\rangle \in V$.

Relation to spectral decomposition Every normal matrix $A \in \mathbb{C}^{n\times n}$ (i.e., $A\,A^\dagger = A^\dagger\,A$) is diagonalizable by an orthonormal basis, that is, there exist a unitary $U \in \mathbb{C}^{n\times n}$ and eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ such that

$$A = U \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} U^\dagger = \sum_{j=1}^{n} \lambda_j\, |u_j\rangle\langle u_j|$$

where the $|u_j\rangle$ are the column vectors of U: $U = \big(u_1|u_2|\cdots|u_n\big)$.

Some eigenvalues can coincide, for example $\lambda_1 = \lambda_2$; then

$$\lambda_1\, |u_1\rangle\langle u_1| + \lambda_2\, |u_2\rangle\langle u_2| = \lambda_1\, P,$$

with $P = |u_1\rangle\langle u_1| + |u_2\rangle\langle u_2|$ the projector onto the (two-dimensional) eigenspace corresponding to eigenvalue λ_1. In general, we can write the spectral decomposition as

$$A = \sum_m \lambda_m\, P_m$$

where the sum is over all *distinct* eigenvalues and the P_m are the projectors onto the corresponding eigenspaces.